

Problem 4

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Problem 1 Suppose the line tangent to the graph of f at $x = 1$ is $y = 6x - 7$. Find an equation of the line tangent to the following curves at $x = 1$:

(a) $y = g(x) = 5(f(x))^4$

Explanation. First, in the equation of the tangent line to $f(x)$ at $x = 1$, when $x = 1$ we have that $y = 6(1) - 7 = -1$. Thus $f(1) = -1$ and therefore $g(1) = 5(-1)^4 = 5$. Hence, a point on our line is $(1, 5)$. Also, since the slope of the given tangent line is $m = 6$, we know that $f'(1) = 6$.

Now, by the chain rule we have that $g'(x) = 5 \cdot 4(f(x))^3 \cdot f'(x)$. So $g'(1) = 20(f(1))^3 \cdot f'(1) = 20(-1)^3(6) = -120$. Thus, the equation of the line tangent to the graph of $y = g(x)$ at $x = 1$ is

$$y - 5 = -120(x - 1)$$

$$y = -120x + 125$$

(b) $y = h(x) = x^2(f(x^3))$

Explanation. $h(1) = 1^2 \cdot f(1^3) = f(1) = -1$. By the product and chain rules:

$$h'(x) = 2x(f(x^3)) + x^2(f'(x^3) \cdot 3x^2) = 2x(f(x^3)) + 3x^4(f'(x^3))$$

Thus, $h'(1) = 2f(1) + 3f'(1) = 2(-1) + 3(6) = -2 + 18 = 16$. So, the equation of the line tangent to the graph of h at $x = 1$ is

$$y - (-1) = 16(x - 1)$$

$$y = 16x - 17$$